



Department of Computer Science and Business Systems

Academic Year 2024– 2025 (Odd/ ~~Even~~ Semester)

Degree, Semester & Branch: B.Tech, V & CSBS

Course Code & Title: CCS335 & Cloud Computing

Name of the Faculty member(s): Dr.M.Gomathy Nayagam

Innovative Practice Description

- **Unit/Topic:** Unit I / Cloud Deployment Model.

- **Course Outcome:** CO1

- **Topic Learning Outcome:** 1b

- **Activity Chosen:** Case Study

- **Justification:**

Using case studies in active learning for the topic of **Cloud Deployment Models** offers numerous benefits, including promoting real-world application, critical thinking, and deep learning. This approach encourages student engagement, enhances communication and collaboration, and helps students develop the practical skills needed to evaluate and select cloud deployment models effectively. By applying theoretical knowledge to real-world situations, students not only gain a deeper understanding of the topic but also prepare themselves for challenges they may encounter in their careers.

- **Time Allotted for the Activity:**20 Minutes

- **Details of the Implementation:**

- Course instructor explained the various cloud deployment models within 25 minutes viz private, public, hybrid and community cloud model with the real time use cases/ case studies SBI (Private Cloud), Netflix (Public Cloud), BBC (Hybrid Cloud)
- Followed by the discussion, course instructor posted the following real time examples/ use cases/ case study to the students for identifying the deployment model for the particular case study and asked the randomly chosen group of students to justify their answers for another 20 minutes
- The case study questions are:
 1. A small business with 25–30 employees has decided to move its data and working functionalities onto the cloud. They are looking for extremely accessible data, easy data backup, cost-efficient. They also need accounting software, customer service tools. Which deployment model is best suited for the considered business?

2. A national bank requires cloud deployment services with high security, privacy, and reliability, allowing only authorized persons to access resources. Which deployment model is best suited for these requirements?
3. A retail company has websites that require high performance at all times. They have on premise servers to handle the working, but sometimes during seasons of sales, they experience periods of spikes in traffic. Which deployment model will handle their traffic spikes and also provide the on premise model?
4. A global company has decided to offer video streaming solutions for business to share their ideas and presentations and pitch them to clients. They wish to adopt cloud scaling to enable larger bandwidth and speed. Which deployment model is best suited for the considered global company?
5. An organization wants to build infrastructure designed for users to access books by members through electronic devices. The borrowed books can be verified using the cloud by sharing resources between devices. Which deployment model will help them build such a system?

- Fig shows the sample case study discussion of the students Ms.Rumana Nachiyar, Ms.S.Varshini and Ms.Sri Janani.

• **CO-PO/ PSO mapping:**

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO 1	3	3	3	2	1	-	-	-	1	1	1	1	3	-	3
(1- Low			2-Moderate			3 - High)									

• **PO/ PSO mapped:**

Innovative practice	PO1	PO2	PO3	PO4	PO5
	3	3	3	2	1
Justification for correlation	The students could be able to apply the basic engineering knowledge to use the various cloud	The students could be able use the cloud deployment model to solve the various issues of real time application.	The students could able to provide the solution for identifying the relationship between the various	The students could be able to identify for which cloud deployment model is suitable to solve complex problems.	The students can be able to use AWS/GCP/ Alibaba Cloud platform for the various deployment model.

	deployment models.		cloud deployment model		
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Innovative practice	PO9	PO10	PO11	PO12	PSO1	PSO3
	1	1	1	1	3	3
Justification for correlation	The students can be able to use SAAS or PAAS or IAAS individually.	The students could be able to discuss with their peer members regarding the various deployment model.	The students could be able to use the cloud deployment model to deploy their project	The students could be able to continuously learn about the deployment model of the cloud,	The students could be able to know the cloud deployment models.	The students could be able to apply the knowledge of cloud deployment model to solve the industry related projects.

- **Images/Screen shot of the practice:**



Fig. Class Listen the Case study Question



Fig. Ms.Ruana Nachiyar, Ms. S.Varishini and Ms. Sri Janani discussed the Case Study Question Answer.

- **Reflective Critique:**

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❖ ***Feedback of practice from students and other stakeholders:***

- The case study activity was incredibly insightful. It allowed the students to apply theoretical knowledge to real-world scenarios, making the concepts of cloud deployment models much clearer. Working in groups helped them to see different perspectives, and the discussions were engaging. Students particularly liked the opportunity to research and analyze real-world challenges, which deepened their understanding. However, some parts were challenging, and they had more time to explore each model in-depth. Overall, it was a valuable experience that improved the student's critical thinking and decision-making skills related to cloud computing deployment model.

❖ ***Benefit of the Practice (E.g. Outcome Attainment would have increased due to innovative practice over conventional practice)***

- It helps students to know, recap and apply all the deployment model in real world scenario and they performed well about the concept during their internal assessment test.

❖ ***Challenges faced in implementation:***

- Make all students to participate in the discussion group.

Signature of Faculty Member
(Dr.M.Gomathy Nayagam)

HOD